Logistics

Q: Why care about this course?

- We will learn fundamental tools of math: logic, set theory, proofs, etc.
- Learn to think abstractly. How much frustration can you tolerate?
- Keep trying, don't isolate yourself.

Syllabus highlights:

- Textbook is free.
- Hw assignments due on Friday 10:00 a.m. Not this week'. Total of 8 assignments. Gradescope.
- Read Hw Guidelines.
- Office hours are awesome!
- LaTeX encouraged.
- Read the textbook before class!

1. Monday 3/25. Statements and conditionals (1.1)

From basic math classes: it's crucial to have precise definitions.
This becomes even more crucial when you want to make proofs.

Definition 1: A **statement** (a.k.a. logical proposition) is a declarative sentence that's either True or false, but not both.

Example 2:

1. "$5x+2$" is not a statement, because it's not a declarative sentence.
2. "$5x+2<14$" is a declarative sentence, but not a statement since we can't evaluate if it's T or F.
3. "There is a real number x such that $5x+2<14$" This is a statement. Happens to be True because $x=0$ does the job: $5\cdot 0+2<14$
4. "$(x+y)^2 = x^2+y^2$" not a statement, can't evaluate if True or False.

5. "For all real numbers x and y, $(x+y)^2 = x^2+y^2$" is a statement
Happens to be False, Take $x=1$, $y=1$ get $(x+y)^2 = 4$ but $x^2+y^2=2$.

6. "There exist real numbers x and y such that $(x+y)^2 = x^2+y^2$" is a statement.
Happens to be True, Take $x=0$, $y=0$

Big Q: How do we decide if a statement is True or False?

Generally, To establish a statement is
- True ⇝ write a mathematical proof.
- False ⇝ give a counterexample (e.g. Example 5 above)

Q: How do we come up with either a proof or a counterexample?

Exploration! Q
- ⟶ Examples
- ⟶ Intuition from previous knowledge
- ⟶ Guess and check (conjectures)
- ⟶ Brainstorming

Definition 3: An integer n is **even** if there exists an integer k so that $n = 2k$.
An integer m is **odd** if there exists an integer ℓ so that $m = 2\ell + 1$.

Proposition 4: If n is an even integer and m is any integer, then nm is even.

Proof: Since n is even, we have $n = 2k$ for some integer k. Then

$$nm = (2k)m = 2(km).$$

Since km is an integer such that $nm = 2(km)$, this shows that nm is even. □ qed

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- Properly punctuated and capitalized.
- Complete English sentences that one can read.
- Not just a string of equations.